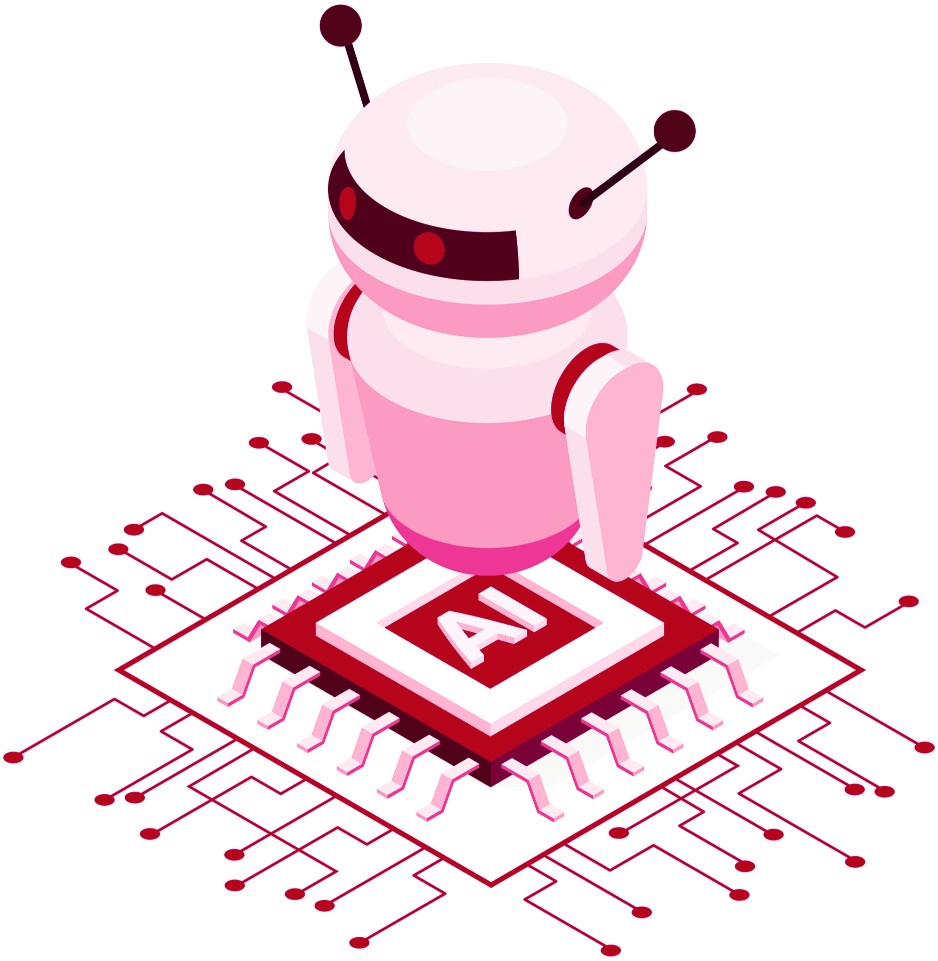


# Deep Reinforcement Learning

Professor Mohammad Hossein Rohban

Solution for Homework 11:

## Imitation Learning and Inverse RL

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# Contents

# 1 Distribution Shift and Performance Bounds

## 1.1 Task 1: Distribution Shift Bound

Show that the total variation distance between state distributions induced by the learned policy and the expert satisfies:

$$\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \le 2T\varepsilon.$$

Given the imitation error bound:

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_{s_t \sim p_{\pi^*}(s_t)}[\pi_\theta(a_t \ne \pi^*(s_t) \mid s_t)] \le \varepsilon$$

Define the per-step error: $e_t = \mathbb{E}_{s_t \sim p_{\pi^*}(s_t)}[\pi_\theta(a_t \ne \pi^*(s_t) \mid s_t)]$ so $\sum_{t=1}^{T} e_t \le T\varepsilon$.

The state distribution shift at time $t$ can be bounded by considering the probability of disagreement events. Let $D_t$ be the event that $\pi_\theta$ and $\pi^*$ disagree on at least one action before time $t$. Using the union bound:

$$P(D_t) \le \sum_{k=1}^{t-1} P(\text{disagree at } k) \le \sum_{k=1}^{t-1} e_k$$

The total variation distance is bounded by the probability of divergence:

$$\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \le 2P(\text{trajectories diverge by } t) \le 2P(D_t)$$

Thus:

$$\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \le 2\sum_{k=1}^{t-1} e_k \le 2\sum_{k=1}^{T} e_k \le 2T\varepsilon$$

## 1.2 Task 2: Return Gap for Terminal Rewards

Assume that the reward is only received at the final step (i.e., $r(s_t) = 0$ for all $t < T$). Show that:

$$J(\pi^*) - J(\pi_\theta) = \mathcal{O}(T\varepsilon).$$

Assume $r(s_t) = 0$ for $t < T$ and $r(s_T) \in [-R_{\max}, R_{\max}]$. The returns are:

$$J(\pi) = \mathbb{E}_{s_T \sim p_\pi(s_T)}\left[r(s_T)\right].$$

The performance gap is:

$$J(\pi^*) - J(\pi_\theta) = \mathbb{E}_{p_{\pi^*}(s_T)}\left[r(s_T)\right] - \mathbb{E}_{p_{\pi_\theta}(s_T)}\left[r(s_T)\right].$$

Bounding this difference:

$$|J(\pi^*) - J(\pi_\theta)| \le \sum_{s_T} |r(s_T)| \cdot |p_{\pi^*}(s_T) - p_{\pi_\theta}(s_T)| \le R_{\max} \sum_{s_T} |p_{\pi^*}(s_T) - p_{\pi_\theta}(s_T)|.$$

From Task 1 (at $t = T$):

$$\sum_{s_T} |p_{\pi^*}(s_T) - p_{\pi_\theta}(s_T)| \le 2T\varepsilon.$$

Thus:

$$|J(\pi^*) - J(\pi_\theta)| \le 2R_{\max}T\varepsilon = \mathcal{O}(T\varepsilon).$$

## 1.3 Task 3: Return Gap for General Rewards

For a general reward function (i.e., $r(s_t) \neq 0$ for arbitrary $t$), show that:

$$J(\pi^*) - J(\pi_\theta) = \mathcal{O}(T^2\varepsilon).$$

For general rewards $r(s_t) \in [-R_{\max}, R_{\max}]$, the return is:

$$J(\pi) = \sum_{t=1}^{T} \mathbb{E}_{s_t \sim p_\pi(s_t)} \left[ r(s_t) \right].$$

The performance gap decomposes as:

$$J(\pi^*) - J(\pi_\theta) = \sum_{t=1}^{T} \left( \mathbb{E}_{p_{\pi^*}(s_t)} \left[ r(s_t) \right] - \mathbb{E}_{p_{\pi_\theta}(s_t)} \left[ r(s_t) \right] \right).$$

Each term satisfies:

$$\left| \mathbb{E}_{p_{\pi^*}(s_t)} \left[ r(s_t) \right] - \mathbb{E}_{p_{\pi_\theta}(s_t)} \left[ r(s_t) \right] \right| \leq R_{\max} \sum_{s_t} \left| p_{\pi^*}(s_t) - p_{\pi_\theta}(s_t) \right|.$$

From Task 1, $\sum_{s_t} |p_{\pi^*}(s_t) - p_{\pi_\theta}(s_t)| \leq 2T\varepsilon$ for all $t$. Thus:

$$|J(\pi^*) - J(\pi_\theta)| \leq \sum_{t=1}^{T} R_{\max} \cdot 2T\varepsilon = 2R_{\max}T^2\varepsilon = \mathcal{O}(T^2\varepsilon).$$

# References

[1] Cover image designed by freepik